

Core Strength Training Can Alter Neuromuscular and Biomechanical Risk Factors for Anterior Cruciate Ligament Injury

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Background: Core stability is influential in the incidence of lower extremity injuries, including anterior cruciate ligament (ACL) injuries, but the effects of core strength training on the risk for ACL injury remain unclear.

Hypothesis: Core muscle strength training increases the knee flexion angle, hamstring to quadriceps (H:Q) coactivation ratio, and vastus medialis to vastus lateralis (VM:VL) muscle activation ratio, as well as decreases the hip adduction, knee valgus, and tibial internal rotation angles.

Study Design: Controlled laboratory study.

Methods: A total of 48 male participants were recruited and randomly assigned to either the intervention group ($n = 32$) or the control group ($n = 16$). Three-dimensional trunk, hip, knee, and ankle kinematic data and muscle activations of selected trunk and lower extremity muscles were obtained while the participants performed side-step cutting. The core endurance scores were measured before and after training. Two-way analyses of variance were conducted for each dependent variable to determine the effects of 10 weeks of core strength training.

Results: The trunk endurance scores in the intervention group significantly increased after training ($P < .05$ for all comparisons). The intervention group showed decreased knee valgus ($P = .038$) and hip adduction angles ($P = .032$) but increased trunk flexion angle ($P = .018$), rectus abdominis to erector spinae coactivation ratio ($P = .047$), H:Q coactivation ratio ($P = .021$), and VM:VL activation ratio ($P = .016$). In addition, the knee valgus angle at initial contact was negatively correlated with the VM:VL activation ratio in the precontact phase ($R^2 = 0.188$; $P < .001$) but was positively correlated with the hip adduction angle ($R^2 = 0.120$; $P < .005$). No statistically significant differences were observed in the trunk endurance scores, kinematics, and muscle activations for the control group.

Conclusion: Core strength training altered the motor control strategies and joint kinematics for the trunk and the lower extremity by increasing the trunk flexion angle, VM:VL activation ratio, and H:Q activation ratio and reducing the knee valgus and hip adduction angles.

Clinical Relevance: Training core muscles can modify the biomechanics associated with ACL injuries in a side-step cutting task; thus, core strength training might be considered in ACL injury prevention programs to alter the lower extremity alignment in the frontal plane and muscle activations during sports-related tasks.

Keywords: core stability; injury prevention; kinematics; muscle activation; trunk; lower extremity

Anterior cruciate ligament (ACL) injuries are common during sports activities. More than 70% of ACL injuries occur in a noncontact situation, and these noncontact ACL injuries primarily result from poor control, which places the athlete's knee joint in an "at-risk" position.^{20,44} In particular, the ability to control the core is important during sports activities, as it provides a stable foundation for the

movement of distal segments.^{24,25} These core muscles, which include the muscles of the trunk and pelvis,³⁰ are known to be activated before the prime movement of the extremities to provide proximal stability for distal mobility.^{25,30} For these reasons, core stability is important and has become a major interest in sports for injury prevention.¹

Since core stability has been shown to be important in lower extremity injuries, core muscle strengthening techniques have been widely used in injury prevention and rehabilitation.^{35,53} Previous studies have suggested that core stability may be beneficial for reducing the risk for ACL injury.^{21,22,24,57,60} Hewett et al^{21,22,24} reported that

neuromuscular control of the trunk can affect the risk for ACL injury; thus, deficits in core stability may be related to an increased risk of injury to the ACL. A 3-year prospective study demonstrated that athletes with worse core stability were more likely to sustain an ACL tear.⁶⁰ In addition, participants with lower core strength were less able to resist hip internal rotation moments, which resulted in excessive knee valgus movement during weightbearing exercise.⁵⁷ The literature above suggests core muscle strengthening may help reduce the risk for ACL injury in athletes through alteration of biomechanical risk factors associated with noncontact ACL injury.

Although previous studies have suggested that core stability is influential in the incidence of lower extremity injuries, the connection between core strengthening and the risk for ACL injury remains unclear. While the core muscles do not cross the knee joint, core stability can modulate the lower extremity alignment and loads during dynamic tasks.³⁸ Thus, it is worthwhile to investigate the relationship between core strengthening and at-risk biomechanical profiles. However, to the best of the authors' knowledge, no longitudinal studies have investigated how core strength training affects biomechanical ACL injury risk factors such as the knee flexion, valgus, and tibial internal rotation angles^{23,33}, hip adduction angle¹⁵, hamstring to quadriceps (H:Q) muscle coactivation ratio⁵; or medial to lateral quadriceps activation ratio³⁹ during the side-step cutting maneuver. Therefore, the purpose of this study was to examine the effect of core muscle strengthening on lower extremity joint kinematics and muscle activations of selected trunk and lower extremity muscles during side-step cutting. It was hypothesized that muscle strength-trained participants would demonstrate decreased hip adduction, knee valgus, and tibial internal rotation angles, as well as an increased knee flexion angle, H:Q muscle coactivation ratio, and vastus medialis to vastus lateralis (VM:VL) muscle activation ratio.

METHODS

Participants

An a priori power analysis based on the data in the pilot study was performed using G*power software Version 3.1.9.2. Effect sizes of 0.90, 0.75, and 0.68 in the knee flexion, valgus, and hip adduction angles, respectively, were used to determine the sample sizes. The sample sizes ($n = 10$ based on knee flexion data, $n = 13$ based on knee valgus data, and $n = 15$ based on hip adduction data) were calculated to obtain a statistically significant effect, with an alpha level of .05 and a desired power of 0.80.

A total of 58 recreationally active men were randomly assigned to either the strength training intervention group ($n = 34$; age, 22.4 ± 2.6 years; height, 175.7 ± 2.9 cm; mass, 72.3 ± 8.1 kg; body mass index [BMI], 23.4 ± 2.6) or the matched control group ($n = 24$; age, 23.1 ± 1.2 years; height, 173.8 ± 3.6 cm; mass, 66.6 ± 5.1 kg; BMI, 22.1 ± 1.5) using the random number generator in an Excel spreadsheet program. If the random number was ≤ 0.50 , the participant was assigned to the intervention group; otherwise, the participant was assigned to the control group, and the investigators were blinded by not being informed of the group allocation. Participants were excluded if they had a lower extremity and/or trunk injury experienced in the past 6 months that prevented participation in physical activity for more than 2 weeks, or if they had a history of ACL injuries or other musculoskeletal injuries requiring surgery. Before participation, all of the men were asked to sign an informed consent form approved by the institutional review board. A total of 32 participants in the intervention group (age, 22.9 ± 2.4 years; height, 175.1 ± 2.8 cm; mass, 72.2 ± 8.3 kg; BMI, 23.6 ± 2.7) and 16 participants in the control group (age, 23.1 ± 1.1 years; height, 174.0 ± 4.0 cm; mass, 67.7 ± 6.6 kg; BMI, 22.3 ± 1.7) successfully completed the study and were included in the data analyses. The 10 men who did not complete the study dropped out for private reasons, such as scheduling conflicts, refusal to participate, and injuries.

Experimental Protocol

Participants in the intervention group conducted exercises of core muscles for 10 weeks (Table 1). The program began within 1 week of the pretraining experiments and was performed 3 days a week, on alternating days. The men in the training intervention group were required to participate in at least 27 of the 30 sessions (90%) for inclusion in the final data set, and the training programs were led by certified sports trainers. The number of sets and repetitions in each set were recorded for each training program. The muscle contraction pace was controlled to standardize the rate of contraction across participants. In the first 5 weeks, participants held the prone- and side-plank and bridge position with 2 legs for 60 seconds. Other exercises were employed at 1 repetition for 5 seconds: contraction for 1 second, holding of the contracted position for 2 seconds, return to the starting position for 2 seconds, and then rest for 60 seconds. In the last 5 weeks, participants held the same position with 1 leg for 30 seconds and then switched legs. Each repetition of the other core training exercises was employed for 1 second, followed by rest for 20 seconds.

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TABLE 1
Core Strength Training Program

Program	Weeks 1-5	Weeks 6-10
Warm-up, 1.3-km jogging	15 min	15 min
Core training		
1. Leg raise	• 1 set:	• 1 set:
2. Crunch	- 12 repetitions	- 20 repetitions
3. Superman	- For 60 s	- For 20 s
4. Plank hip twist	• Total 3 sets	• Total 5 sets
	• 60 s rest	• 20 s rest
5. Prone-plank	• 1 set:	• 1 set:
6. Side-plank (both sides)	- Hold for 60 s	- Hold for 30 s with each limb only
7. Supine bridge	• Total 3 sets	• Total 3 sets
	• 60 s rest	• 60 s rest
Stretches		
1. Quadriceps	30 s each	30 s each
2. Hamstring		
3. Calf stretches		
4. Latissimus dorsi		
5. Hip muscles		
6. Pectorals/biceps		

The experimental task employed in this study was a side-step cutting maneuver, not only because the ability to cut quickly and effectively is critical to overall performance in sports, but also because cutting tasks account for 57% of the noncontact ACL injuries in collegiate basketball and soccer.^{18,43} Participants were instructed to run 3 m at a pace of 3.5 to 4.0 m/s before contacting their dominant foot on the force plate and then changing their direction to the opposite limb. Limb dominance was determined by asking which leg they could kick a ball farthest with, and all of participants indicated the right leg as their dominant leg in this study. The line was marked on the floor to make participants cut at an angle of 45° in the direction of progression. A digital metronome was used to provide an auditory cue for the participant to maintain the desired approach speed. Before the actual experimental trials, each participant in the intervention or control group was given approximately 15 minutes to walk and run to get used to the provided shoes,⁵² and each participant was instructed to perform several practice trials to become familiar with the procedures and instrumentation. If the foot was not facing straight ahead or was placed outside of the force plate, then the corresponding trials were discarded. To remove the effect of footwear, all participants wore the same running shoes: Nike Downshifter 6 (NIKE Inc). All participants in both the intervention and control groups were instructed to wear the provided shoes 3 days a week.

A 3-dimensional motion capture system equipped with 10 infrared cameras (9 Eagle and 1 Raptor; Motion Analysis Corp) was used to record the motion of the trunk, hip, knee, and ankle joint at a sampling rate of 400 Hz.³² Reflective markers (ie, 12.5-mm diameter spheres) were placed on the anatomic bony landmarks: left and right acromion, sternum, right scapula, bilateral anterior superior iliac spines, sacrum, greater trochanter, midpoint of the femur, lateral and medial epicondyles of the femur, lateral and medial

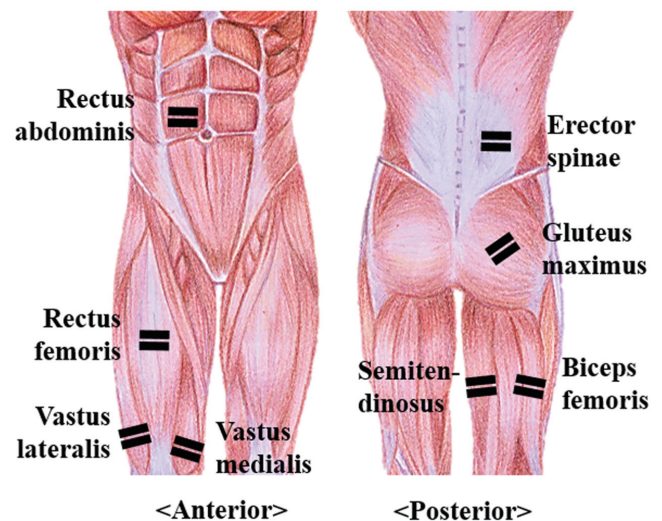


Figure 1. Electrodes on the muscle bellies of core and lower extremity.

plateaus of the tibia, midpoint of the tibia, lateral and medial malleoli, calcaneus, and first and fifth metatarsal heads.^{2,26} An electromyography (EMG) system (LXM5308; Laxtha Inc) was used to record muscle activation from the rectus abdominis, erector spinae, gluteus maximus, rectus femoris, vastus medialis or lateralis, semitendinosus, and biceps femoris with a sampling rate of 1024 Hz during side-step cutting. Surface electrodes were attached to the muscle bellies with an interelectrode distance of 20 mm in recommended locations (Figure 1). To test the core stability, prone- and side-plank endurance tests were used.¹³ Less expensive testing in the form of isometric trunk endurance testing has been popular and practical for assessing core endurance; thus, it

has been used in previous studies as a field test to evaluate core stabilizer endurance.^{36,47} The objective of these endurance tests was to hold a static position for as long as possible, and the tests were terminated when the participant was no longer able to hold the position.

Data Analysis

Each trial was defined as the period from the initial heel strike to the toe-off, as determined by the force plate recordings. The initial contact of side-step cutting was identified by finding the first frame at which the vertical ground-reaction force exceeded 20 N. The measured kinematic data were filtered using a zero-lag fourth-order Butterworth low-pass filter at a cutoff frequency of 10 Hz. To calculate the joint kinematics, the coordinate systems for each body segment were defined following previous methods.^{2,26}

Raw EMG data were filtered with a band-pass Butterworth filter at 20 to 500 Hz.¹⁰ Each trial of EMG data was fully rectified¹⁰ and smoothed with root mean square processed using a moving window and time constant of 15 milliseconds.⁴² The processed EMG was then normalized to the EMG amplitudes obtained from maximal voluntary isometric contractions (MVCs) during manual muscle tests. The mean EMG amplitudes (% MVC) for each muscle during the precontact and loading phases were calculated. The precontact phase was defined as a 50-millisecond time period before heel strikes.^{5,19} The loading phase was defined as the initial 50% of the stance phase during the side-step cutting. This time period of deceleration has been associated with noncontact ACL injuries.^{8,19} The VM:VL activation ratio was calculated as the average EMG amplitude of the vastus medialis divided by the average EMG amplitude of the vastus lateralis during the precontact and loading phases. The rectus abdominis to erector spinae (RA:ES) coactivation ratio was calculated as the average EMG amplitude of the rectus abdominis divided by the average EMG amplitude of the erector spinae. The H:Q coactivation ratio was calculated as the average EMG amplitude of the hamstrings (biceps femoris and semitendinosus) divided by the average EMG amplitude of the quadriceps (rectus femoris and vastus medialis or lateralis).¹¹

Statistical Analysis

The 2 successful trials were averaged individually and then averaged as a group to generate mean values and standard deviations. A 2×2 (group $\times$ time) analysis of variance was conducted for each dependent variable. Post hoc paired *t* tests with false discovery rate (FDR) correction were performed to compare each dependent variable between pre- and posttraining in each group if a significant interaction effect of group $\times$ time was detected. The Benjamini-Hochberg procedure⁶ was chosen to control the FDR of 5% among the variables, which showed significant group $\times$ time interaction effects. Changes in core stability and kinematic variables were calculated by subtracting the measured values at pretraining from the measured values at posttraining. To

determine the relationship between the core stability differences and the changes in significant kinematic variables found from post hoc tests, Pearson correlation coefficient was used. All statistical analyses were performed using MATLAB Version R2018a (The Mathworks Inc).

RESULTS

No group $\times$ time interactions were observed for the approach speed, nor were significant main effects for group or time present (intervention group: pretraining, 3.74 ± 0.37 m/s; posttraining, 3.76 ± 0.42 m/s; control group: pretraining, 3.60 ± 0.42 m/s; posttraining, 3.57 ± 0.28 m/s). Thus, the statistical comparisons presented here were considered to be not influenced by the approach speed.

The endurance scores significantly increased after training in the intervention group ($P < .05$ for all comparisons) (Table 2) as determined by post hoc paired *t* tests with FDR correction, while no differences were found in the control group. There were no significant differences in either the intervention group or the control group for the quadriceps and hamstring muscle strength ($P > .05$ for all comparisons) (Table 2).

Significant group $\times$ time interaction effects were observed for the trunk flexion, hip adduction, and knee valgus angles at initial contact ($P < .05$ for all parameters) (Table 3), while there were no significant differences for peak values of corresponding parameters. The trunk flexion angle at initial contact increased by 19% after core training ($P = .018$) (Table 3). After training, the hip adduction and knee valgus angles at initial contact significantly decreased by 47% and 46%, respectively, compared with the corresponding values in pretraining ($P = .032$ and $P = .038$, respectively) (Table 3). The change in hip adduction was negatively correlated with the change in core endurance in the side-plank position ($R^2 = 0.216$; $P = .007$), but it was not correlated with the change in core endurance in the prone-plank position ($R^2 = 0.017$; $P = .474$) (Figure 2). The change in knee valgus angle was negatively correlated with the changes in both the prone-plank score ($R^2 = 0.225$; $P = .006$) and the side-plank score ($R^2 = 0.278$; $P = .002$) (Figure 2). No significant correlations between the change in other kinematic parameters and the change in core endurance were found.

Significant group $\times$ time interaction effects were also observed for the muscle activation amplitudes of the rectus abdominis, gluteus maximus, rectus femoris, vastus medialis, and semitendinosus, as well as for the VM:VL, RA:ES, and H:Q activation ratios in the precontact phase ($P < .05$ for all parameters) (Figure 3). Post hoc paired *t* tests with FDR correction showed that there were increases in the rectus abdominis activation ($P = .030$), vastus medialis activation ($P = .013$), semitendinosus activation ($P = .032$), VM:VL activation ratio ($P = .016$), RA:ES coactivation ratio ($P = .047$), and H:Q coactivation ratio ($P = .021$) and decreases in the rectus femoris activation ($P = .001$) after core strength training (Figure 3). The contribution of VM increased after training (pretraining, $31.2\% \pm$

TABLE 2
Prone- and Side-Plank Endurance Scores and Concentric Isokinetic Quadriceps/Hamstring
Muscle Strength in Both the Training and Control Groups^a

	Training Group		Control Group	
	Pretraining	Posttraining	Pretraining	Posttraining
Prone-plank, min ^b	3.00 ± 1.20	3.73 ± 1.50 ^c	2.95 ± 1.10	2.89 ± 1.21
Side-plank, min ^b	1.31 ± 0.32	1.65 ± 0.39 ^c	1.18 ± 0.37	1.19 ± 0.35
Quadriceps, % (N·m/kg)	285.1 ± 56.8	293.4 ± 51.8	272.4 ± 38.9	267.7 ± 42.7
Hamstrings, % (N·m/kg)	134.9 ± 24.7	135.9 ± 29.4	137.9 ± 29.1	134.8 ± 28.2
H:Q strength ratio	0.48 ± 0.09	0.47 ± 0.08	0.51 ± 0.11	0.51 ± 0.12

^aData are presented as mean ± SD. H:Q, hamstring to quadriceps.

^bSignificant interaction effects ($P < .05$).

^cSignificant differences between pre- and posttraining with post hoc t test (Benjamini-Hochberg corrected $P < .05$).

TABLE 3
3-Dimensional Kinematics of Trunk, Hip, Knee, and Ankle in Both Training Intervention and Control Groups^a

	Training Group		Control Group	
	Pretraining	Posttraining	Pretraining	Posttraining
At initial contact				
Trunk flexion ^{b,c}	14.4 ± 7.6	17.1 ± 6.0 ^d	11.2 ± 4.4	10.8 ± 4.1
Trunk right-side tilt	8.4 ± 3.7	8.1 ± 4.9	9.8 ± 4.7	10.3 ± 5.0
Trunk right-side rotation	10.4 ± 5.5	11.8 ± 7.8	12.8 ± 5.4	12.3 ± 6.2
Hip flexion	42.8 ± 6.1	44.0 ± 6.3	40.3 ± 6.1	42.0 ± 5.3
Hip adduction ^c	4.3 ± 6.2	2.3 ± 6.7 ^d	3.2 ± 5.6	5.0 ± 6.6
Hip internal rotation	4.5 ± 8.2	4.9 ± 7.4	5.1 ± 3.8	4.9 ± 6.2
Knee flexion	29.2 ± 5.4	28.7 ± 6.7	31.3 ± 2.3	30.9 ± 4.1
Knee valgus ^c	1.3 ± 1.6	0.7 ± 1.5 ^d	1.0 ± 4.1	1.6 ± 2.5
Tibial internal rotation	0.9 ± 1.6	1.2 ± 1.6	1.1 ± 1.1	1.2 ± 1.0
Ankle dorsiflexion	3.0 ± 7.5	3.2 ± 6.7	3.3 ± 5.0	3.8 ± 6.3
Ankle inversion	6.3 ± 4.4	6.3 ± 5.2	6.5 ± 4.0	7.0 ± 3.2
Ankle external rotation	7.6 ± 5.6	5.8 ± 7.1	5.6 ± 4.3	5.9 ± 4.4
Peak values				
Peak trunk flexion	20.9 ± 8.9	22.9 ± 8.5	20.6 ± 5.7	20.2 ± 6.5
Peak trunk right-side tilt	14.9 ± 3.6	14.5 ± 4.9	15.6 ± 6.2	16.2 ± 5.1
Peak trunk right-side rotation	5.0 ± 6.1	4.8 ± 6.9	4.2 ± 6.1	4.4 ± 4.8
Peak hip flexion	36.8 ± 7.2	38.8 ± 7.1	35.4 ± 7.4	37.8 ± 6.0
Peak hip adduction	0.6 ± 6.0	0.4 ± 6.5	0.6 ± 1.9	1.5 ± 2.2
Peak hip internal rotation	0.8 ± 8.2	1.1 ± 7.4	0.6 ± 3.9	0.9 ± 8.1
Peak knee flexion	65.4 ± 6.8	64.5 ± 6.2	66.3 ± 6.5	65.8 ± 4.9
Peak knee valgus	1.0 ± 3.1	0.6 ± 3.3	1.1 ± 14.4	0.9 ± 8.7
Peak tibial internal rotation	0.8 ± 2.3	1.5 ± 2.3	0.7 ± 1.8	0.9 ± 3.6
Peak ankle dorsiflexion	11.9 ± 4.5	10.5 ± 5.6	13.5 ± 7.2	12.9 ± 7.5
Peak ankle inversion	5.3 ± 3.2	6.1 ± 4.3	8.0 ± 2.8	8.2 ± 3.1
Peak ankle external rotation	14.8 ± 5.8	13.7 ± 6.6	16.0 ± 6.7	15.0 ± 5.2

^aThe angles are measured in degrees and the data are presented as mean ± SD.

^bSignificant group effects ($P < .05$).

^cSignificant interaction effects ($P < .05$).

^dSignificant differences between pre- and posttraining with post hoc t test (Benjamini-Hochberg corrected $P < .05$).

8.9%; posttraining, 45.9% ± 11.1%; $P < .001$), but the contribution of rectus femoris to quadriceps decreased (pretraining, 29.5% ± 12.3%; posttraining, 17.8% ± 7.6%; $P < .001$). There were no differences in the contributions of semitendinosus and biceps femoris to hamstrings

between pre- and posttraining. No statistically significant differences in the muscle activations were found between the training and control groups at baseline, nor were any differences found between the 2 data collection time points in the control group.

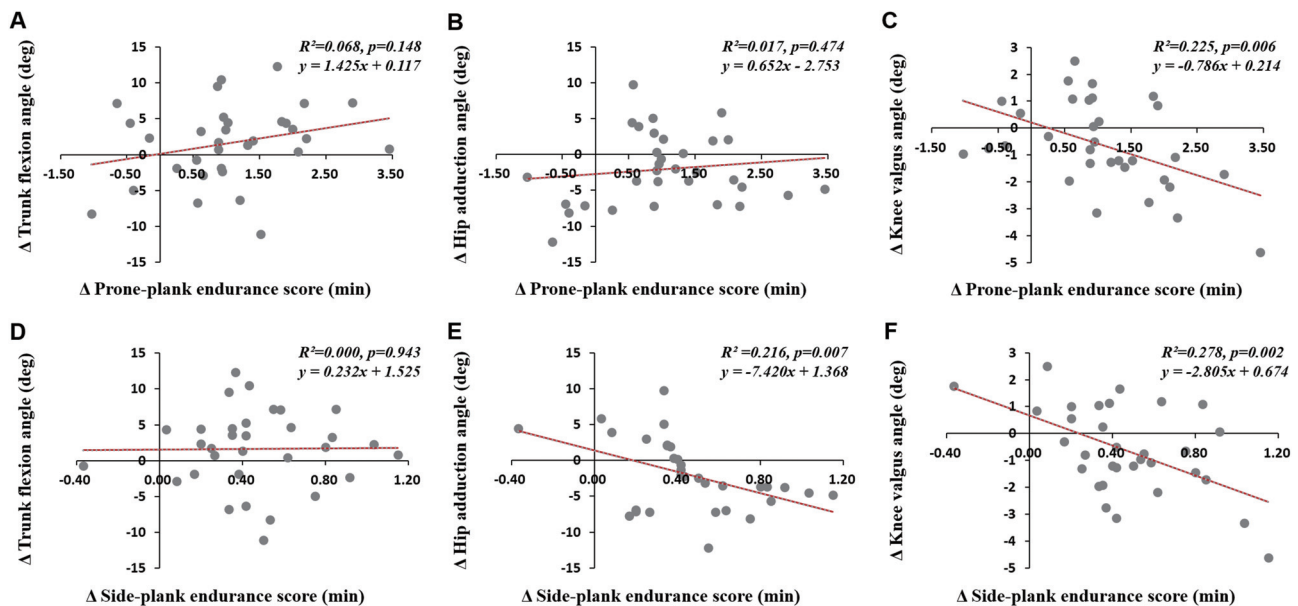


Figure 2. Relationship between change in (A-C) prone-plank endurance score and (D-F) side-plank endurance score and change in angles of the trunk, hip, and knee joint. The change in each variable was calculated by subtracting the pretraining values from the posttraining values. Positive angles are flexion for the trunk, adduction for the hip, and valgus for the knee joint.

DISCUSSION

The main finding of this study was that 10 weeks of core strength training altered the lower extremity alignment at initial contact and muscle activations during the side-step cutting maneuver. As there were no differences for the quadriceps and hamstring muscle strength, the differences in kinematic and muscle activation parameters in the trunk and lower extremity seemed to be the consequences of core strengthening. The results of this study supported our hypothesis that core muscle strengthening would decrease the knee valgus and hip adduction angles and increase the VM:VL activation ratio. These significant changes identified in joint kinematics caused by altered muscle activation patterns provide novel evidence that core strength training can modify an individual's motor control strategy.

The knee valgus angle at initial contact significantly decreased after core muscle strength training in this study. Furthermore, the more the core was strengthened, the smaller the knee valgus angle at initial contact became during the cutting task (Figure 2). Our findings agree with the observations reported by Willson et al,⁵⁷ as core stability was associated with knee valgus collapse. These authors reported that participants who had greater isometric core strength showed lower knee valgus angles during a single-leg squat. Although several studies have suggested that dynamic knee valgus is not a major mechanism for ACL injury,^{15,17,55,59} other studies have identified an association between the dynamic knee valgus angle and noncontact ACL injury risk.^{3,23,50} With respect to the risk for injury, the larger knee valgus moment and angle observed in women compared with men are thought to be

predictors of ACL injuries.^{14,37,51} This biomechanical difference between the sexes is also in accordance with the greater incidence rate of ACL injuries in women. Taken together, the current findings indicate that strengthening the core helps athletes avoid knee valgus alignment at initial contact, and hence may be beneficial for reducing the risk of noncontact ACL injury. An evidence-based future study of the relationship between the changes in the knee valgus angle by core strength training and ACL injury rates needs to be conducted.

Our results show that the VM:VL activation ratio during the precontact phase significantly increased in the intervention group and was negatively correlated with the knee valgus angle at initial contact ($R^2 = 0.188$; $P < .001$) (Figure 4). This means that an increased VM:VL activation ratio after core strength training likely decreases the knee valgus angle at initial contact by changing the neuromuscular control strategy before foot contact (Figure 4). It has been previously documented that women tend to demonstrate VL dominance during side-step cutting, in contrast, to male athletes who tend to show VM activation dominance.^{4,39} This difference between sexes in muscle activation patterns may increase the potential for a valgus knee position.³⁹ The low medial to lateral vastii muscle activation ratio can cause the lateral part of the knee joint to be compressed and the medial joint to be opened, which can cause knee valgus alignment.³⁹ The current study demonstrates how core muscle strength training may reduce the knee valgus angle.

Core strength training altered the VM:VL activation ratio during the precontact phase in this study. It is unclear how core muscle strengthening leads to an increase in the VM:VL activation ratio. However, it is

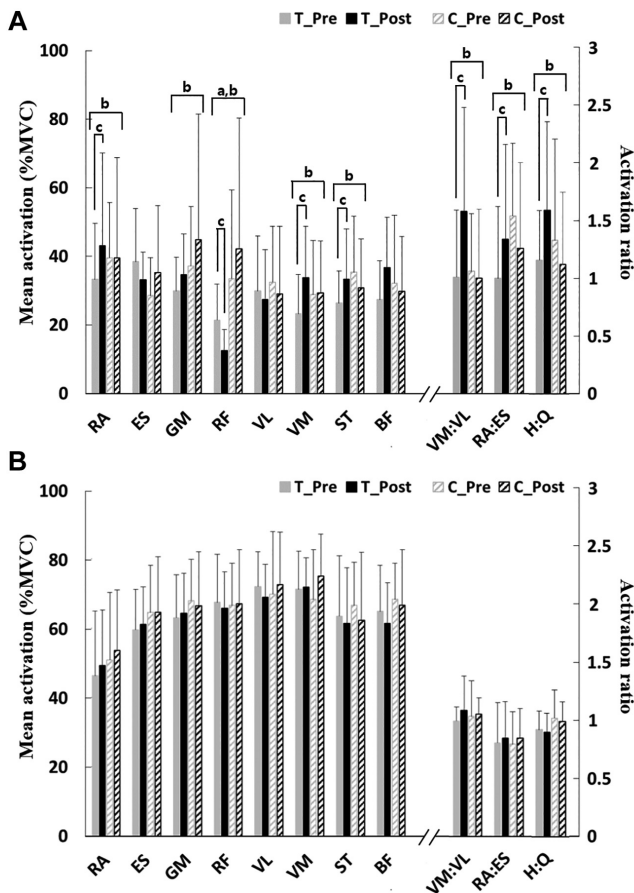


Figure 3. Mean muscle activation amplitude (% MVC) and SD during the (A) precontact phase and the (B) loading phase for both the training intervention and control groups. ^aSignificant group effects ($P < .05$); ^bsignificant interaction effects ($P < .05$); ^csignificant differences between pre- and posttraining with post hoc t test (Benjamini-Hochberg corrected $P < .05$). RA, rectus abdominis; ES, erector spinae; GM, gluteus maximus; RF, rectus femoris; VL, vastus lateralis; VM, vastus medialis; ST, semitendinosus; BF, biceps femoris; H, hamstrings; Q, quadriceps; T_Pre, pretraining for training group; T_Post, posttraining for training group; C_Pre, pretraining for control group; C_Post, posttraining for control group.

plausible that the preparatory activation strategy contributing to the knee valgus angle is altered by core strength training to protect the knee joint. It has been suggested that the preparatory motor control strategy can be modified by training, thereby promoting dynamic joint stability.¹² A more stable core allows more efficient distal segment movements, which protect the distal joint,^{29,30} so taken together, the core strength training examined in this study may alter the muscle activation strategy by increasing the VM:VL activation ratio, which leads to reducing the knee valgus angle (Figure 4). Therefore, our results suggest that core strengthening induces appropriate neuromuscular adaptations in the medial and lateral quadriceps, which leads to more neutral coronal alignment or prevents knee joint valgus collapse.

The hip adduction angle at initial contact decreased after core muscle strength training, and this result indicates that strengthening the core muscles helps reduce the risk of ACL injury in athletes by restraining their knee valgus angle. Considering that an increased hip adduction angle is associated with an increased risk of ACL injuries,¹⁵ strengthening core muscles can decrease the risk of ACL injury by controlling the hip kinematics. Although it is not directly linked to ACL injury incidence, hip adduction has been cited as a significant predictor of knee valgus.²⁸ As an excessive hip adduction angle during landing can cause the knee joint to move medially and increase the knee valgus angle,^{45,46} it is possible that the incidence of knee injuries associated with excessive knee valgus angles can be reduced if excessive hip adduction during weightbearing activities such as cutting or landing is prevented. This concept is consistent with our results, which showed that there was a positive correlation between the hip adduction angle and the knee valgus angle at initial contact ($R^2 = 0.120$; $P = .005$) (Figure 4). Previous studies reported that strengthening the gluteal muscles prevents hip adduction and internal rotation during single-leg support,⁵⁸ and stronger hip strength was associated with less knee valgus angle during functional task.⁵⁴ Thus, a reduced hip adduction angle with a strengthened gluteal muscle through core strength training would contribute to reducing the knee valgus angle.

Our results also show that the trunk flexion angle and RA:ES coactivation ratio, as well as the H:Q coactivation ratio, significantly increased after training. These changes can help increase dynamic stability, thereby playing an important role in reducing the risk of ACL injury. A previous video analysis has shown that athletes who sustained an ACL injury were leaning backward at the time of injury.⁴⁸ Blackburn and Padua⁷ also reported that a more flexed position of the trunk during landing decreased the landing force and quadriceps muscle activation level. Similarly, participants who extended their trunk implemented a quadriceps-dominant strategy.⁴⁹ These studies indicate that when the trunk is more extended, the quadriceps are more recruited in an attempt to maintain balance. In addition, there is a biomechanical association between the hamstrings and the trunk in dynamic movements; the force generated by the hamstring muscles is greater in trunk flexion than in trunk extension.³¹ Navacchia et al⁴¹ reported that an increase in hamstring contractions provided a protective effect on the ACL by reducing the peak anterior tibial shear force. In addition, when examining the magnitude of agonist-antagonist muscle coactivation during risky movements, women showed greater quadriceps muscle activations and relatively lower hamstring muscle activations, thereby showing a lower H:Q ratio of muscle activation compared with men in the precontact phase when they were performing side-step cutting or landing.^{5,19,40} This neuromuscular difference between sexes is also in agreement with the higher incidence rate of ACL injuries in women. The trunk flexion in the sagittal plane can be decreased by the co-contraction of the erector spinae muscles, which increases spine stiffness.²⁹ Taken together, the increased trunk

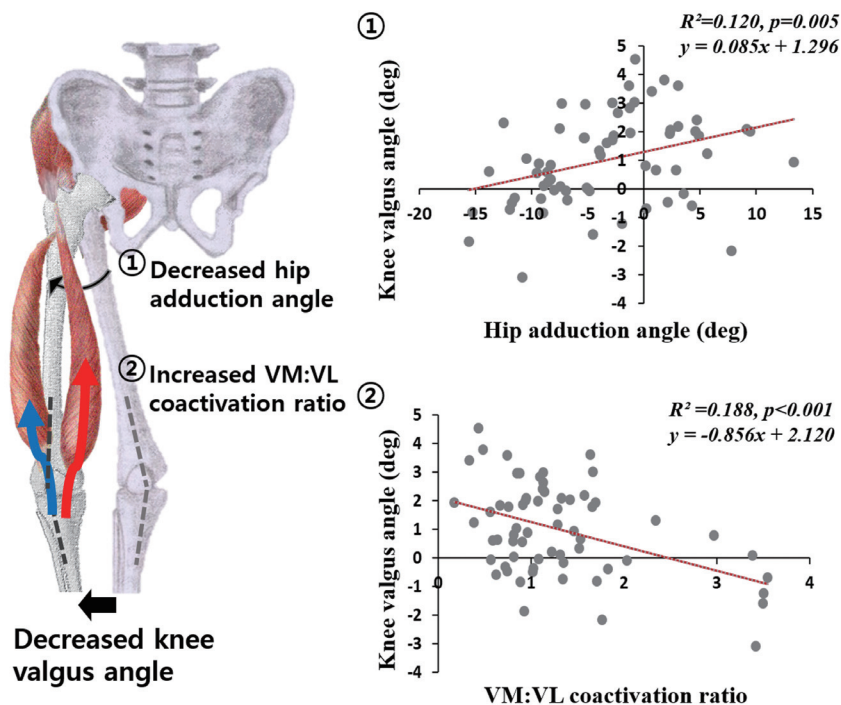


Figure 4. Relationship between the knee valgus angle at initial contact with the hip adduction angle at initial contact and the vastus medialis to vastus lateralis (VM:VL) activation ratio during the precontact phase in both pre- and posttraining. Positive values indicate the knee valgus and hip adduction angles, respectively.

flexion with the increased H:Q coactivation ratio after core muscle training can be beneficial for reducing the risk for noncontact ACL injuries.

There are some limitations to this study. First, the term “core stability” is used in this study, but only core endurance was measured. Core stability is instantaneous, and efficient functioning requires the successful integration of adequate muscular characteristics defined by strength and endurance and sensorimotor control.^{9,34} Thus, additional measurements of core muscle strength are recommended to gain a deeper understanding of the effect of core strengthening, although some previous studies have demonstrated that muscle endurance may be a more important factor than muscle strength in core stability because it helps core muscles maintain an efficient trunk position during prolonged exercise.^{9,16,27} In addition, the current study evaluated the effect of a core strength training intervention on joint kinematics and muscle activation patterns only for ACL injury prevention, and the actual incidence of ACL injuries in participants after the completion of training was not tracked. Future studies on the effect of core strength training on the incidence of ACL injuries are needed to evaluate whether the observed changes in kinematics and muscle activation patterns by core strength training actually resulted in decreased ACL injury rates. Furthermore, because only male participants were included in this study, the current results cannot be generalized to women. However, according to a recent meta-analysis of ACL injury reduction training programs,

there is insufficient information to make conclusions regarding the effectiveness of ACL injury training prevention programs in men compared with women.⁵⁶ Therefore, this study can provide additional evidence supporting the effectiveness of core strengthening as an ACL injury prevention program in male participants.

CONCLUSION

This study provides evidence that core strength training alters the motor control strategy and at-risk biomechanical parameters associated with an ACL injury during the cutting maneuver by reducing the knee valgus and hip adduction angles and increasing the VM:VL and H:Q activation ratio. Thus, training the core muscles might be considered in ACL injury prevention programs to alter the lower extremity alignment in the frontal plane and muscle activations during sports-related tasks.

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